MPG No. 22.1.79 $\bar{6}$ [Type I, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & M_z \end{bmatrix}$$

$$M_z = \mathbb{M}_z^{(1)}$$

* Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & M_{zzz} \\ 0 & M_{xxz} & 0 \\ M_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = \mathbb{M}_z^{(1)} - \mathbb{M}_z^{\alpha(1)}$$
$$M_{zzz} = 3\mathbb{M}_z^{(1)} + 2\mathbb{M}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & M_{xxz} \\ 0 & 0 & 0 \\ M_{yzx} & -\frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxz} = 2\mathbb{M}_z^{(2)}$$
$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$
$$\frac{M_{xxz}}{2} = \mathbb{M}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & M_{yzy} & 0 \\ -M_{yzy} & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$
$$M_{yzy} = \mathbb{M}_z^{(3)}$$
$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & 0 & 0 & -M_{xxzx} \\ M_{xxzx} & -M_{xxzx} & 0 & 0 & 0 & M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4s}^{(1)}$$
$$M_{xxzx} = \mathbb{M}_{4c}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & 0 & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & 0 & 0 & -M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = 2\mathbb{T}_{3c}^{(1)}$

$M_{xxzx} = -2\mathbb{T}_{3s}^{(1)}$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & M_{xxzx} & 0 \\ -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxzx} & -M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = \mathbb{T}_{3c}^{(2)}$

$M_{xxzx} = \mathbb{T}_{3s}^{(2)}$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$

$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 & M_{xxzy} & M_{xxxz} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 & -M_{xxzy} & -M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & -M_{xxxz} & M_{xxzy} & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} + \mathbb{T}_{3c}^{(2)}$

$M_{xxzx} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} + \mathbb{T}_{3s}^{(2)}$

$M_{xxzy} = \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} - \mathbb{T}_{3c}^{(2)}$

$M_{xxxz} = \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} - \mathbb{T}_{3s}^{(2)}$

$M_{yzxx} = \mathbb{M}_{4s}^{(1)} - 2\mathbb{T}_{3c}^{(1)}$

$M_{yzxy} = -\mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)}$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$

$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & M_{yyyz} \\ 0 & 0 & 0 \\ -M_{xxxz} & M_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyz} \\ 0 & 0 & 0 \\ M_{xxxz} & -M_{yyyz} & 0 \\ 0 & 0 & -M_{xxxz} \\ -M_{yyyz} & -M_{xxxz} & 0 \end{bmatrix}$$

$M_{xxxz} = \mathbb{M}_{4c}^{(1)}$

$M_{yyyz} = -\mathbb{M}_{4s}^{(1)}$